

SARTA vs. Line-by-Line Models (LBLRTM)

To appreciate how SARTA operates, it helps to compare it directly to a **Line-by-Line Radiative Transfer Model** like [LBLRTM](#). [1]

Feature	Line-by-Line (LBLRTM)	SARTA (Fast Parameterization)
Calculation Method	Integrates millions of discrete molecular line profiles using HITRAN database entries.	Evaluates a 10-to-15 term algebraic polynomial matrix for each layer.
Spectral Resolution	Monochromatic (fraction of a wavenumber, e.g., 0.0005 cm ⁻¹).	Channel-convolved (matches the instrument's footprint).
Processing Speed	Minutes to hours per atmospheric profile.	Milliseconds per atmospheric profile.
Primary Use Case	Reference standard, sensor design, and deriving fast-model coefficients.	Operational weather forecasting and real-time climate data ingestion.

Note: SARTA does not replace LBLRTM; it relies on it. LBLRTM serves as the foundational "truth" model used to pre-calculate SARTA's static regression coefficient matrices (\$A_i\$). [2]

Instrument Adaptation: Grating vs. Interferometers

Because SARTA calculates optical depths directly at the sensor channel level rather than monochromatic frequencies, its internal coefficient arrays must be custom-tuned to a satellite's exact optical hardware. [2]

1. AIRS (Atmospheric Infrared Sounder)

- **Hardware Nature:** AIRS is a **grating array spectrometer** with 2,378 independent detectors.
- **SARTA Adjustment:** Because individual detector elements can drift or experience shifts over decades in orbit, SARTA includes localized channel calibration offsets. It handles each of the 2,378 channels as independent entities, mapping dedicated polynomial arrays tailored to each channel's unique Spectral Response Function (SRF). [3, 4]

2. CrIS and IASI (Interferometers)

- **Hardware Nature:** Both CrIS and IASI are **Fourier Transform Spectrometers (FTS)**. They do not split light with a physical grading grating; instead, they generate interferograms that are mathematically transformed into smooth spectra. [4, 5]
- **SARTA Adjustment:** FTS systems produce uniform channel spacing. SARTA takes advantage of this by applying an **un-apodized** framework. It matches the instrument's mathematical self-apodization function directly inside the regression coefficients to handle things like the ringing side-lobes typical of Michelson interferometers. [4]

Operational Role in Climate Science

Because SARTA operates fast enough to process millions of footprints a day, it serves as the core "observation operator" for major remote sensing architectures like NASA's CLIMCAPS processing pipeline. [6]

By continuously evaluating its rapid polynomial sets against incoming satellite views, it allows researchers to instantly isolate greenhouse gas anomalies, global temperature trends, and moisture variations spanning multiple decades without clogging supercomputing clusters. [7, 8]

Would you like to examine how SARTA handles **instrument noise modeling (NEAT)** within its validation schemes, or look at how its **adjoint model code** is utilized during operational atmospheric profile retrievals?

[1] <https://rtweb.aer.com>

[2] <https://par.nsf.gov>

[3] <https://agupubs.onlinelibrary.wiley.com>

[4] <https://airs.jpl.nasa.gov>

[5] <https://www.sciencedirect.com>

[6] <https://events.ecmwf.int>

[7] <https://ntrs.nasa.gov>

[8] <https://agupubs.onlinelibrary.wiley.com>